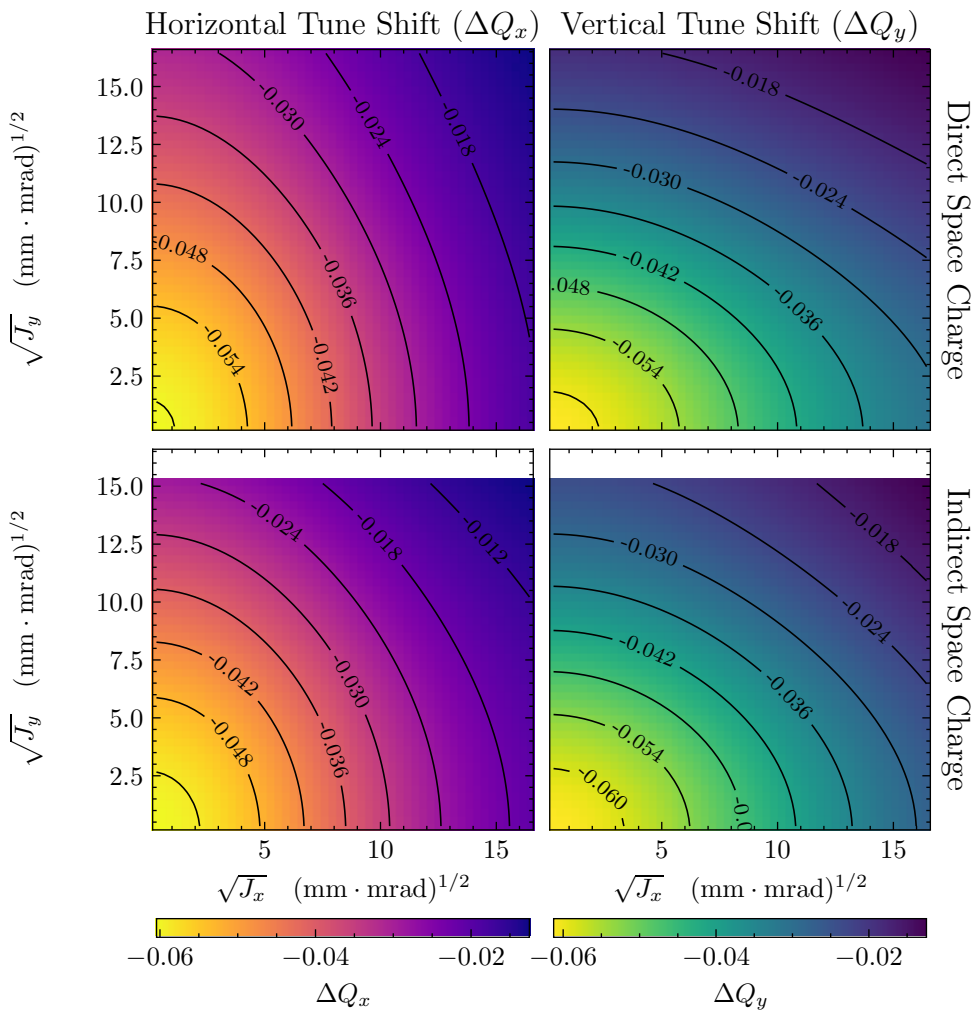
Horizontal Tune Shift (ΔQ_x)
Vertical Tune Shift (ΔQ_y)
Direct Space Charge
Indirect Space Charge
$\sqrt{J_y}$ (mm · mrad)$^{1/2}$
$\sqrt{J_x}$ (mm · mrad)$^{1/2}$
15.0
12.5
10.0
7.5
5.0
2.5
5
10
15
-0.018
-0.024
-0.030
-0.036
-0.042
-0.048
-0.054
-0.060
-0.012
−0.06
−0.04
−0.02
ΔQ_x
ΔQ_y